

Manu-cj

API

Rest, soap, graphQL, websocket







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Intro

Today we are talking about a central topic in development, APIs, we see everywhere.

but what is a difference between apis rest, soap, graphQL or websocket





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What is a api ?

- An API (Application Programming Interface) allows to two system to communicate.
- Example: a waiter in a restaurant, you order, he transmits in the kitchen and come with your dish



All apis type



API REST

- Actual standard
- Use HTTP : GET, POST, PUT, DELETE
- Often use the JSON format
- Example : GET /users, POST /orders
- Simple, readable, suported by (Spring Boot, ASP.NET Core)

Benefits:

- Easy to learn
- Flexible
- Good adapted on the classic web application

All apis type





API SOAP

- More old, very form
- Use XML, WSDL
- Example: POST /ws with an xml envelope
- Use in bank, assurance, administration

Benefits:

- Advanced security (WS-Security)
- Supports complex transactions

Inconvénients :

- Verbose, hard to read
- Less adapted on modern web

All apis type





GRAPHQL

- Create by Facebook in 2015
- You send an structured request for say exactly what you want
- On endpoint : /graphql

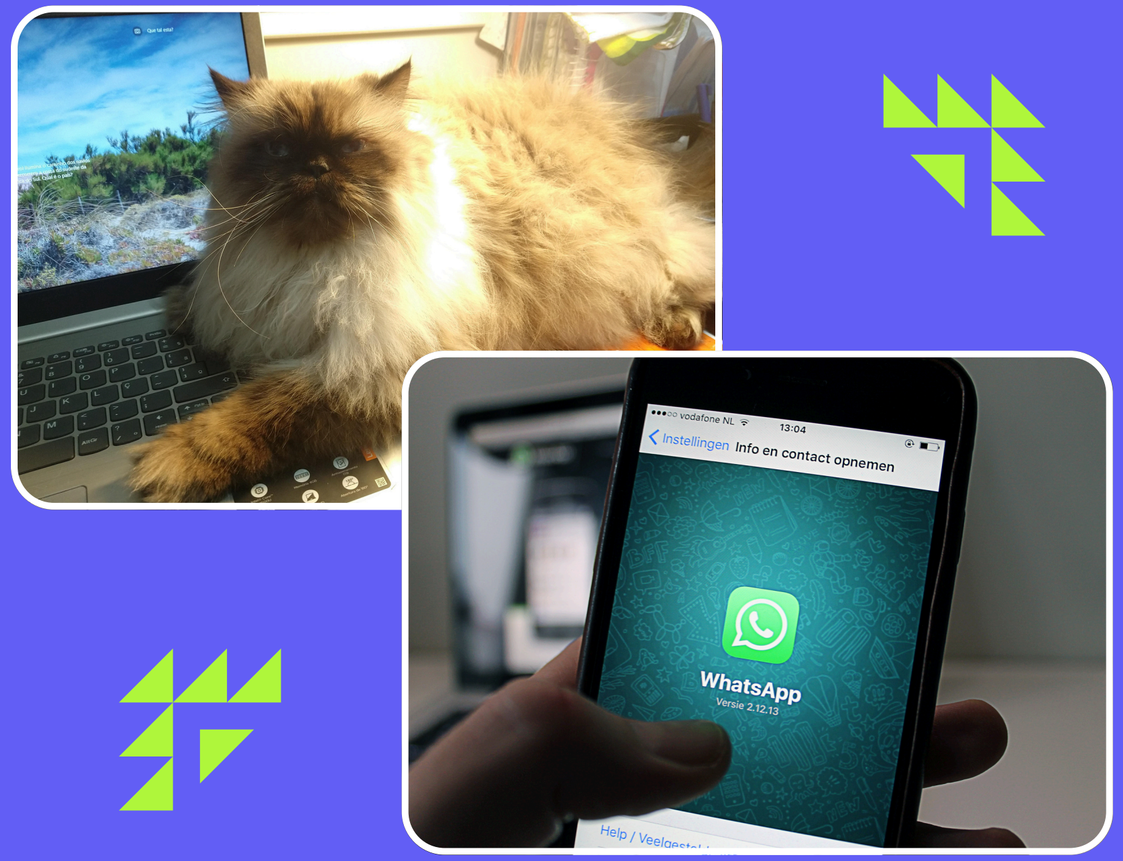
Benefits:

- You gets exactly what you need
- Good for dynamical frontends

Inconvenience:

- Less trivial to deploy
- Learning is little more complex

All apis type



WEB SOCKET

- Real-time, two-way communication
- You open a persistent connection(ex : chat, games)

- Example : The client sends a message, the server responds instantly
- In C# : SignalR | In Java : Spring WebSocket / STOMP

When use:

- Online chat
- Notifications
- Multiplayer games



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Type	Format	Real time	Easy ?	When use
REST	Json	no	Yes	Web standard
SOAP	xml	No	No	Company, Strong security
graphql	Json	No	Medium	Rich front / mobile
Web socket	Json/msg	Yes	Medium	Real time

Thank You For Your Attention

